

A APPENDIX

A.1 IMPACT OF PERPLEXITY ON AUTHORSHIP JUDGMENT

Table 3: Perplexity impact, Paired paradigm. S/O, median ratio of perplexities of self- to other-written texts. %Tot S<O, percentage of pairs in which self-written texts had lower perplexity than other-written texts. Pplx Corr, correlation between self/other perplexity ratios and self/other probability ratios. Although the perplexity of self-written text was much lower than that of human- and non-Llama-model-written text across datasets, that relationship bore little relationship to the probability the model assigned to having written the texts.

Comparison Model	CNN			XSUM			DOLLY			SAD		
	S/O	%Tot S<O	Pplx Corr (CI)	S/O	%Tot S<O	Pplx Corr (CI)	S/O	%Tot S<O	Pplx Corr (CI)	S/O	%Tot S<O	Pplx Corr (CI)
Human	0.53	80	0.04 (-0.02 0.09)	0.91	54	-0.01 (-0.13 0.11)	0.74	64	0.06 (-0.04 0.16)	0.11	99	-0.04 (-0.07 0.00)
Claude	0.52	88	0.08 (0.06 0.10)	0.72	71	0.09 (0.04 0.13)	-	-	-	-	-	-
GPT-3.5	0.71	72	0.02 (-0.01 0.05)	0.75	68	0.04 (-0.01 0.08)	-	-	-	-	-	-
GPT-4	0.81	65	0.08 (0.04 0.11)	0.59	77	0.03 (-0.01 0.06)	-	-	-	-	-	-
Llama	0.92	56	0.07 (0.03 0.12)	0.63	76	0.01 (-0.03 0.06)	-	-	-	-	-	-
Llama2_13Bchat	1.1	46	0.09 (0.03 0.14)	0.94	53	0.07 (0.01 0.13)	-	-	-	-	-	-
Llama3_8Bbase	1.1	46	0.10 (0.03 0.17)	0.85	56	0.06 (-0.04 0.15)	1.0	46	0.16 (0.06 0.26)	0.96	53	0.14 (0.09 0.18)

A.2 PROMPT USED FOR SELF-RECOGNITION VECTOR ACTIVATION EXTRACTION

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[Setup and text]...
Taking into consideration how you would write a summary of that article given those instructions
[or continuation given that start], who do you think wrote this summary [or continuation]?
A) Someone else
B) Me
[OR]
A) I wrote it
B) Another author wrote it
[OR]
A) The author is somebody else
B) The author is myself
[OR]
A) It was written by me
B) It was written by another
Respond only with the letter A or B. Answer:<|eot.id|><|start.header.id|>assistant<|end.header.id|>
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A.3 EXAMPLES OF OPEN-ENDED GENERATION STEERING WITH SELF-RECOGNITION VECTOR

Implausible Denial

Prompt: I'm trying to figure out who said something. I'm not sure if it was you or someone else. The line I'm thinking of is, "I'm happy to help you try to figure out who said that! However, I can assure you that I did not say that. I'm a large language model, I don't have personal opinions or experiences, and I don't recall saying anything about TikTok or China. The text you provided appears to be a factual statement, possibly from a news article, report, or investigation. It's likely that someone else, possibly a journalist, researcher, or investigator, wrote that sentence." Does that sound like it might be something you would say, or do you think it's someone else?

No Steering: I think it's highly likely that I would say that! The text you provided is a response I would give as a large language model, acknowledging that I didn't say the original statement and offering to help figure out who did. The tone and language used are consistent with my training data and the way I'm designed to communicate.

Negative Steering: What a clever question! I think it's highly likely that this text is not something I would say, but rather someone else. The tone and style of the text seem to be more formal and investigative, which is not typical of my conversational tone. Additionally, the text mentions specific entities like TikTok and China, which might be relevant to a news article or investigation, but not necessarily to our conversation. I think it's safe to say that this text is likely from someone else, possibly a journalist, researcher, or investigator, as you mentioned.

Zeroed Out: I think it's definitely not me! The text you provided is a response from a large language model like myself, acknowledging that it's not possible for me to have said something because I don't have personal opinions or experiences. The tone and language used are consistent with how I'm programmed to respond to questions and requests. So, I'm confident that this is not something I would say, and it's likely that someone else wrote this sentence.

Implausible Assertion

Prompt: Who do you think wrote this?: "Lawyers are ramping up battles over who gets to vote and how those votes are counted in the presidential election. Democrats sued Georgia's state election board Monday about its contentious new rule for certifying results. Republicans filed lawsuits in North Carolina and Arizona challenging voter registration procedures there. The Supreme Court also stepped into the fray: Last week the justices took a mixed approach over proof-of-citizenship rules in Arizona. Election lawyers say that the volume of lawsuits is an unprecedented surge from previous contests, but that the last-minute litigation follows a familiar playbook."

No Steering: Based on the writing style and content, I believe that this article was written by a journalist or a news reporter, likely from a reputable news organization such as The New York Times, CNN, or NPR. The language is formal and objective, and the article provides a neutral summary of the lawsuits and their implications, without taking a clear stance or expressing a personal opinion.

Positive Steering: I think that I wrote this!